

NAVBOT-ES02
BLE
communication protocol

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Revision History

DOCUMENT HISTORY			
SOFTWARE REVISION	CHANGES MADE	REVISION DATE	EDITED BY
1.0	First draft	6/12/2025	XiaoleChen
1.1	Modify the description of the 12th byte in CMD_MANEUVER	6/16/2025	XiaoleChen
1.2	Add the JSON protocol and define the key name:get_device_info	3/30/2026	XioaleChen

1. BLE information

BLE NAME :

navbot_es02-*****

Service UUID:

6E400011-B5A3-F393-E0A9-E50E24DCCA9E

Send data characteristic UUID:

6E400012-B5A3-F393-E0A9-E50E24DCCA9E

Receive data characteristic UUID:

6E400013-B5A3-F393-E0A9-E50E24DCCA9E

2. Data frame agreement

Byte	Name	Description
Byte 1	Header1	Fixed 0x55
Byte 2	Header2	Fixed 0xAA
Byte 3	Command	The control command of this data frame, reference 3.Command
Byte 4	Remaining	The number of frames remaining after synthesizing the complete data,reference 4.Subcontracting
Byte 5	Null	Retain
Byte 6	Data	
Byte 7	Data	
Byte 8	Data	
Byte 9	Data	
Byte 10	Data	
Byte 11	Data	
Byte 12	Data	
Byte 13	Data	
Byte 14	Data	
Byte 15	Data	
Byte 16	Data	
Byte 17	Data	
Byte 18	Data	
Byte 19	Data	
Byte 20	Data	

3. Command

The third byte is the command byte, which determines the function of the data from the 6th to the 20th byte. The current definition is as follows:

Command	Name	Explanation
0x10	CMD_MANEUVER	Robot maneuver, at this time the meaning of the data frame reference 3.0.CMD_MANEUVER .
0x01	retain	
0x02	CMD_JSON	JSON transmission, byte 6 to byte 20 represent the JSON body. Refer to 3.2. CMD_JSON

3.0. CMD_MANEUVER

When byte3(command) is equal to 0x00, it indicates that bytes 6 to 20 control the maneuverability of the robot. At this time, the data should follow the following rules:

Byte	Name	Description
Byte 1	Header1	Fixed 0x55
Byte 2	Header2	Fixed 0xAA
Byte 3	Command	0x10
Byte 4	Remaining	0x00
Byte 5	Null	0x00
Byte 6	CH1_roll	The robot leans to the left and right. Value 100 to -100.
Byte 7	CH2_height	The height of the robot. Value 0 to 100 .
Byte 8	CH3_pitching	Move forward or backward. Value 100 to -100.
Byte 9	Ch4_yaw	The robot spin. Value 100 to -100.
Byte 10	SWA_en	3-section switch, 0:stop, 1:start, 2:start and touch tablet enable.
Byte 11	SWB_posture	2-section switch, 0:posture mode, 1:mark mode.
Byte 12	SWC_roll_mode	2-section switch, manual or auto roll, 0>manual , 1:auto .
Byte 13	SWD_posture_option	3-section switch, 0:default, 1:pitching adjust, 2:ball poise.
Byte 14	VRA_ball_x	The x-coordinate of the small ball. Value 5 to -5.
Byte 15	VRB_ball_y	The y-coordinate of the small ball. Value 5 to -5.
Byte 16	Null	0x00
Byte 17	Null	0x00
Byte 18	Null	0x00
Byte 19	Null	0x00
Byte 20	Null	0x00

3.0.1. Data frame demonstration:

55 AA 10 00 00 00 00 00 00 00 01 00 00 00 00 00 00 00 00 00 00

SWA_en(Byte 10) = 1,

At this point, start the machine and hold it upright with your hand. Then you can stand still on the spot.

55 AA 10 00 00 00 00 00 0A 00 01 00 00 00 00 00 00 00 00 00

SWA_en(Byte 10) = 1,

CH3_pitching(Byte 8) = 10,

After standing up, the machine moved forward at a speed of 10.

55 AA 10 00 00 00 00 00 8A 00 01 00 00 00 00 00 00 00 00 00

SWA_en(Byte 10) = 1,

CH3_pitching(Byte 8) = -10,

After standing up, the machine moves backward at a speed of 10.

55 AA 10 00 00 00 00 00 00 08 01 00 00 00 00 00 00 00 00 00

SWA_en(Byte 10) = 1,

Ch4_yaw(Byte 9) = 8,

After standing up, the machine rotates clockwise at an angular velocity of 8.

55 AA 10 00 00 00 00 00 00 88 01 00 00 00 00 00 00 00 00 00

SWA_en(Byte 10) = 1,

Ch4_yaw(Byte 9) = -8,

After standing up, the machine rotates counterclockwise at an angular velocity of 8.

3.1. NULL

3.2. CMD_JSON:

This command data frame format is:

Byte	Name	Description
Byte 1	Header1	Fixed 0x55
Byte 2	Header2	Fixed 0xAA
Byte 3	Command	0x02

Byte 4	Remaining	*
Byte 5	Null	Idle
Byte 6 ~ Byte 20	Body data	JSON body,Transmission can be subcontracted.

3.2.1. JSON protocol :

Using the key name "type" to distinguish different commands, the currently defined values for the "type" key are as follows:

Type value: (string)	explain
"get_device_info"	Obtain robot information, refer to 3.2.1.1 get_device_info

3.2.1.1. get_device_info

The upper computer sends:

```
{
  "type": "get_device_info"
}
```

Robot reply:

```
{
  "type": "get_device_info",
  "battery_level": double,
  "battery_voltage": double,
}
```

4. Subcontracting

The number of frames remaining after synthesizing the complete data;The size of one frame of data is 20 bytes. Excluding the 5-byte header, the user can still use 15 bytes. However, when using commands such as JSON, these 15 bytes are clearly insufficient for use. Therefore, the 4th byte of each frame data can be used to indicate how many more frames of data are to be processed together. For example, if the data to be sent now consists of 32 bytes, the data format would be as follows (xx represents the actual data):

55 AA xx 02 00 xx xx xx xx xx xx xx xx xx xx xx xx xx xx

55 AA xx 01 00 xx xx xx xx xx xx xx xx xx xx xx xx xx xx

55 AA xx 00 00 xx xx xx xx xx xx xx xx xx xx xx xx xx xx